E-Learning Platform Sales and Purchase Analysis

Objective

The objective of this project is to design and analyze an e-learning platform database using MySQL to gain insights into learner purchasing behaviour, course performance, category-wise revenue, and overall sales trends.

Problem Statement

An online e-learning platform offers multiple courses across categories such as Technology, Business, Design, and Language. Learners from different countries can purchase one or more courses based on their interests and learning goals.

However, the platform management faces several business challenges:

- The platform lacks visibility into which course categories drive the highest demand and revenue.
- There is no systematic identification of top-performing courses based on total purchases and sales.
- High-value learners contributing the most revenue cannot be identified.
- The platform cannot track which countries show the highest learner engagement and purchase activity.
- Learners who purchase across multiple categories are not easily identifiable.
- Courses with low demand or zero purchases remain unknown.
- Monthly and yearly sales trends are not currently monitored.
- Missing course prices compromise accurate revenue analysis and reporting.
- A deeper understanding of learner purchasing behaviour is needed for targeted campaigns and bundle offers.
- A structured relational database and SQL-based analysis are required to support better business decisions.

To solve these challenges, this project uses MySQL to design a relational database, perform joins and analytical queries, and generate insights about learner spending, course popularity, category performance, and revenue growth.

Dataset Description:

- The dataset consists of purchase transaction records along with supporting dimension tables such as Learners Dimension and Courses Dimension.
- The purchase fact table includes attributes like Purchase ID, Learner ID, Course ID, Quantity, and Purchase Date.
- The dimension tables provide additional descriptive information related to learners (name, country) and courses (name, category, unit price), enabling detailed analysis of purchase performance across learner segments, course categories, and geographic regions.

Table Description

Learners Table Description

Column Name	Description
learner_id	Unique ID for each learner
full_name	Name of the learner
country	Country of the learner

Course Table Description

Column Name	Description
course_id	Unique ID for each course
course_name	Name of the course
category	Course category such as Technology, Business, Design, or Language
unit_price	Price of the course

Purchase Table Description

Column Name	Description
purchase_id	Unique ID for each purchase
learner_id	Learner ID linked to the learners table
course_id	Course ID linked to the courses table
quantity	Number of courses purchased
purchase_date	Date of purchase

Schema Design

Learners Table

- Primary Key: learner_id

Courses Table

- Primary Key: course_id

Purchases Table

- Primary Key: purchase_id
- Foreign Key: learner_id references learners(learner_id)

- Foreign Key: course_id references courses(course_id)

Relationship Between Tables

- One learner can purchase multiple courses
- One course can be purchased by multiple learners
- The purchases table acts as a bridge between learners and courses

Entity Relationship Summary

- learners → purchases = One-to-Many
- courses → purchases = One-to-Many

Table Import Process

The dataset was created and analyzed using MySQL database management system. The following steps were followed to build the data model:

- Created a new database named e_learning_platform_db using CREATE DATABASE statement.
- Designed three tables (learners, courses, purchases) with appropriate data types and constraints.
- Established foreign key relationships between purchases and both learners and courses.
- Inserted sample data covering 30 learners, 11 courses, and 30 purchase records.
- Performed data cleaning and transformation using SQL UPDATE and SELECT statements.
- Executed analytical queries using JOINs, window functions, and subqueries for insights generation.

Data Cleaning & Transformation:

Learner_ID Standardized

- The Learner_ID column was defined as INT with PRIMARY KEY and AUTO_INCREMENT constraints at table creation to ensure unique, sequential, and consistent identification of each learner.

Course_ID Standardized

- The Course_ID column was defined as INT with PRIMARY KEY and AUTO_INCREMENT constraints to ensure unique, sequential, and consistent identification of each course.

Purchase_ID Standardized

- The Purchase_ID column was defined as INT with PRIMARY KEY and AUTO_INCREMENT constraints to ensure unique, sequential, and consistent identification of each purchase transaction.

Country Standardized

- The Country column in the learners table was defined with VARCHAR(100) data type and NOT NULL constraint to ensure complete and consistent country information for every learner.

Course Name Standardized

- The Course_Name column was defined with VARCHAR(100) data type and NOT NULL constraint to ensure every course has a valid, non-empty name for accurate identification.

Category Standardized

- The Category column was defined with VARCHAR(100) data type to classify courses into Technology, Business, Design, or Language categories for meaningful aggregation.

Unit Price Standardized

- The Unit_Price column was defined with DECIMAL(10,2) data type to store monetary values with two decimal places.
- NULL values present in two courses (Data Science and Figma UI/UX Design) were identified and replaced with the average unit price across all courses using an UPDATE statement with subquery.

Formula:

```
UPDATE courses SET unit_price = (SELECT AVG(unit_price) FROM courses)
WHERE unit_price IS NULL;
```

Quantity Standardized

- The Quantity column was defined with INT data type and a CHECK constraint (Quantity > 0) to prevent zero or negative enrollment quantities, ensuring only valid purchase records are stored.

Purchase Date Standardized

- The Purchase_Date column was defined with DATE data type and a DEFAULT constraint set to CURRENT_TIMESTAMP, automatically capturing the current date when new records are inserted without manual input.

Total Spending Calculated

- Total spending per purchase was calculated at query time using the formula (Quantity × Unit_Price).
- The IFNULL() function was used to handle any remaining NULL unit_price values safely.
- The FORMAT() function was applied to display currency values with two decimal places.

Formula:

```
FORMAT(IFNULL(c.unit_price, (SELECT AVG(unit_price) FROM courses)) * p.quantity, 2) AS
total_amount
```

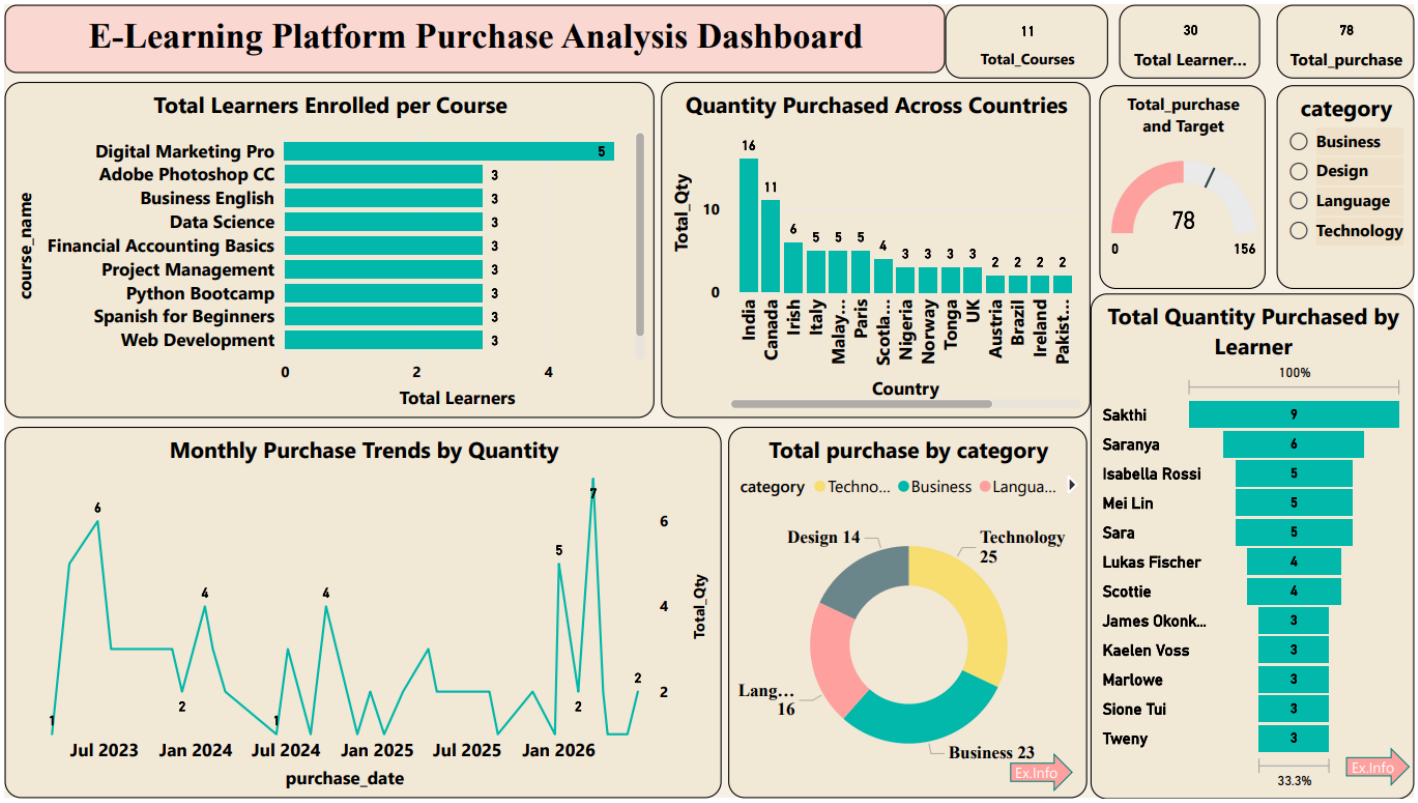
Spender Tier Created

- A Common Table Expression (CTE) named learner_spending was created to compute total spending per learner.
- A CASE statement was then applied to classify learners into High Spender (>5000), Medium Spender (2000–5000), and Low Spender (<2000) tiers for targeted marketing and retention strategies.

Formula:

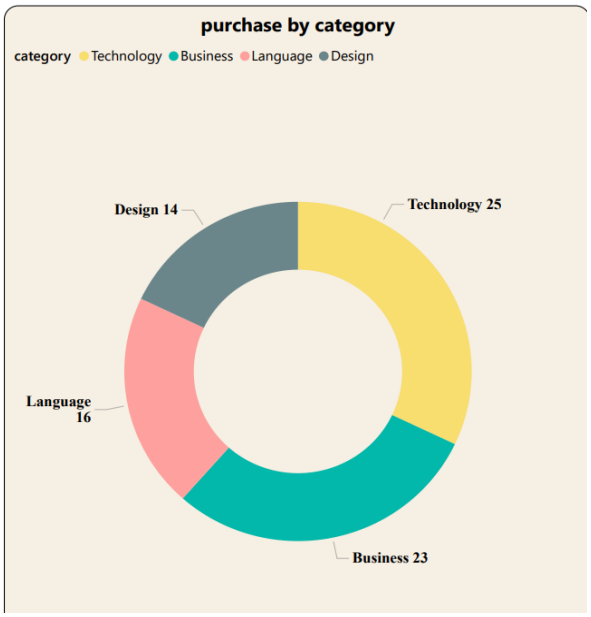
```
CASE
   WHEN Total_Spent > 5000 THEN 'High Spender'
   WHEN Total_Spent >= 2000 AND Total_Spent <= 5000 THEN 'Medium Spender'
   ELSE 'Low Spender'
END AS Spender_Tier
```

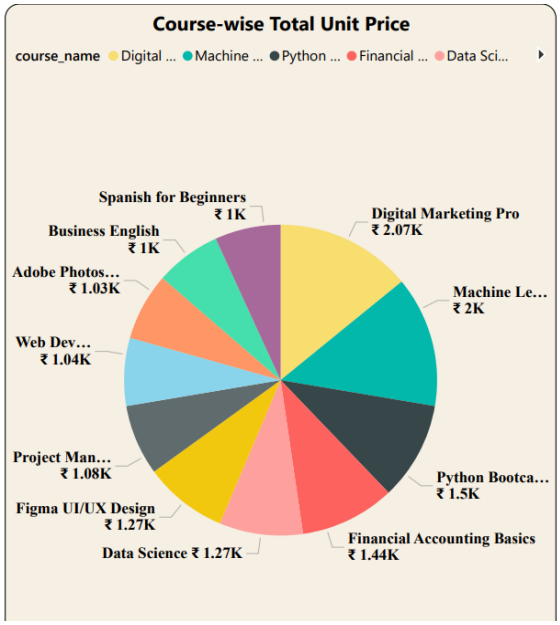
Visualization Analysis



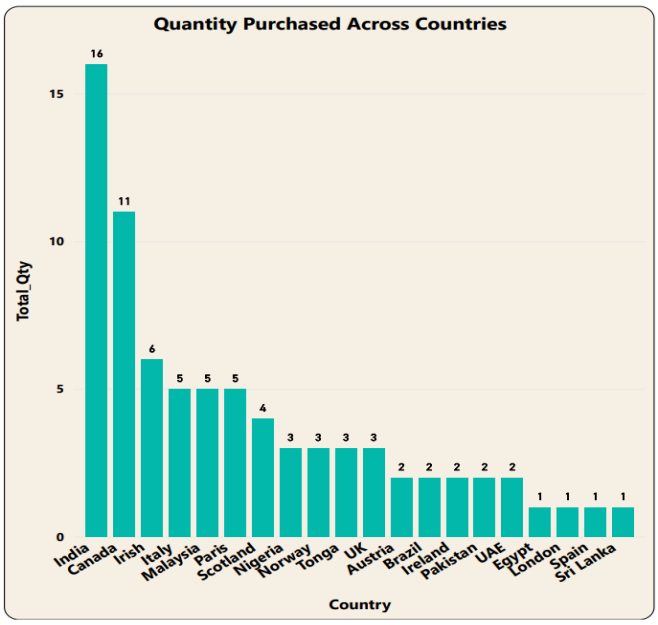
- The platform recorded 78 total purchases across 30 courses and 11 learners.
- Technology category had the highest purchase quantity with 25 enrollments, followed by Business with 23. Language and Design categories had lower demand with 16 and 14 purchases respectively.
- India had the highest purchase quantity with 16 enrollments, followed by Canada with 11 enrollments.
- Sakthi was the top learner with 9 total purchases, while Saranya made 6 purchases.

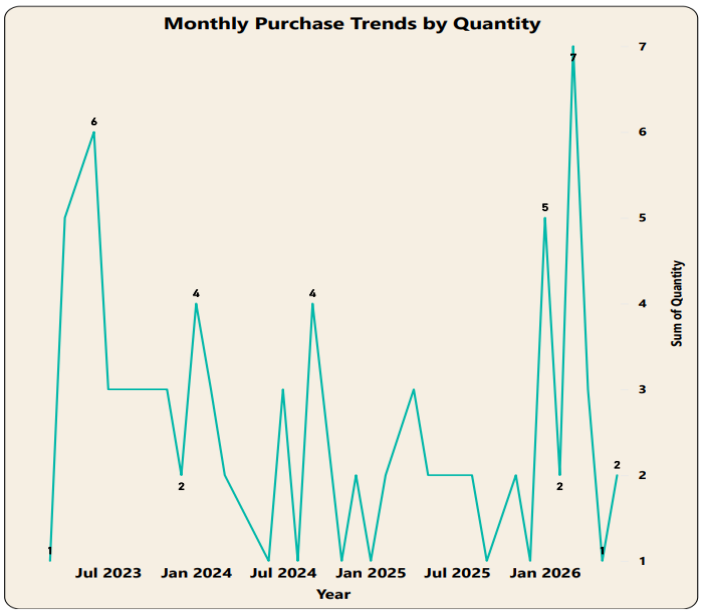
- Digital Marketing Pro had the highest enrollment count with 5 learners. Data Science and Adobe Photoshop CC also performed strongly in purchase quantity.
- Total cumulative sales reached approximately ₹1.03 lakh by 2026.



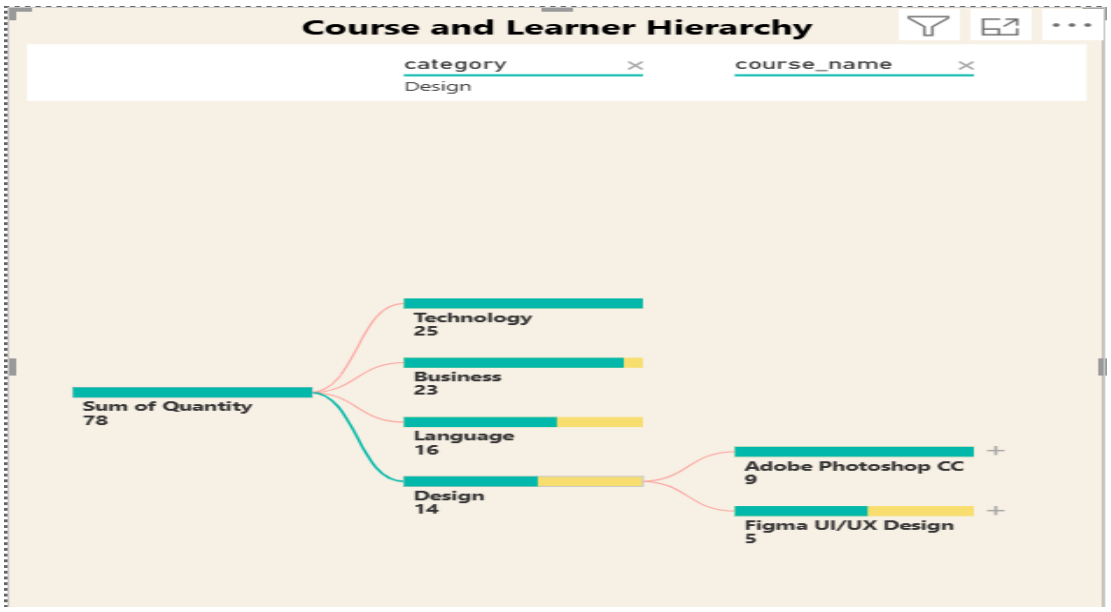


- Technology and Business courses performed better because learners are more interested in career-oriented and high-income skill development programs.
- Digital Marketing Pro generated high enrollments because marketing skills are useful across multiple industries and job roles.
- Higher-priced courses such as Data Science still achieved strong purchase volumes, showing that learners are willing to invest more in courses with better career opportunities.
- Lower-performing courses like Figma UI/UX Design and Project Management may have lower visibility, weaker demand, or less attractive content compared to top-performing courses.
- Sales dropped in 2025 compared to 2024, which may be due to fewer course launches, lower marketing activity, or seasonal decline in learner engagement.





- Technology and Business categories are likely to remain the strongest revenue-generating categories in the future because they currently dominate purchase quantity and learner interest.
- Courses related to digital skills, analytics, and career growth are expected to attract more learners over time.
- Countries such as India and Canada are likely to continue contributing the highest number of enrollments.
- Repeat learners such as Sakthi and Saranya may continue purchasing more courses if loyalty programs or discounts are introduced.
- Total sales are expected to grow further if the platform expands its high-demand course categories and increases marketing efforts.



- Increase the number of Technology and Business courses because these categories generate the highest demand and revenue.
- Promote high-performing courses such as Digital Marketing Pro, Data Science, and Adobe Photoshop CC through advertisements and featured sections on the platform.
- Offer discounts, referral bonuses, and loyalty rewards to repeat learners like Sakthi and Saranya to improve retention.
- Improve low-performing courses like Figma UI/UX Design and Project Management by updating course content, adding certifications, or creating bundle offers.
- Focus marketing campaigns in low-performing countries such as Spain, Sri Lanka, and Egypt to increase global reach and learner enrollments.

Conclusion

- The platform has built a strong foundation with consistent learner demand, generating 78 total purchases and cumulative sales of more than ₹1.03 lakh by 2026.
- Technology and Business categories are the key drivers of platform growth, contributing the highest purchase quantities and showing the strongest learner interest.
- Top-performing courses such as Digital Marketing Pro, Data Science, and Adobe Photoshop CC play a major role in increasing both learner enrollments and revenue.

- A small group of repeat learners, especially Sakthi and Saranya, contributes a significant share of purchases, highlighting the importance of customer retention strategies.
- Countries like India and Canada are currently the strongest markets, while lower-performing countries provide opportunities for future expansion.
- Although sales performance dropped in 2025, the recovery in 2026 shows that the platform has strong growth potential when supported by effective marketing and high-demand course offerings.
- By focusing on high-demand categories, improving underperforming courses, and introducing loyalty programs, the platform can further strengthen learner engagement and increase long-term revenue.